Calibration run of 2 November: the six-point cubic

Crystal **infi2439_SC_1** (SC-cut, temperature PLL on the 6566 kHz mode) · run `qcm_run_2026-11-02_122657` · Mon 2 Nov 13:27 → 20:20 CET (6.9 h) · Peltier, PT100 11.5–64.8 °C · analysis 3 Nov 2026

Summary

- **Your cubic works.** Recomputed over the whole run, where nothing was deposited, the compensated output stays within **7.55 Hz peak to peak** (−3.1 to +4.5 Hz; 52 pm Au, 100 ng/cm²) over 11.5–64.8 °C. That range moves the mass mode by 166 Hz, so the gain is 22×, ramps of up to 6 °C/min included. Between 20 and 40 °C, ramps included, the band is 4.79 Hz (33 pm). The run itself logged with the linear weekend profile, which was 307 Hz off at 64 °C.
- **The largest single error is not thermal:** at 17:11:01 CET (60.1 °C) fM fell **3.80 Hz** in under a second. fT and both amplitudes did not move. Both 64 °C points were taken after the drop, so the cubic absorbed it: above 40 °C it sits up to 3.8 Hz below where the crystal was before. The same six points refitted with the drop taken out (**infi2439_SC_1_stepfree**) are identical below 40 °C (within 0.1 Hz) and roughly halve the error above it.
- **The board temperature acts on the clock again.** During the 64 °C soak the error follows the TSic at 1.98 Hz/°C (0.33 ppm/°C; the weekend gave 0.25–0.4) with 40 mHz rms left, and fT moves 1.16× as much as fM (a clock gives 1.10×). The evening cool-down of the board (0.9 °C) cost 1.7 Hz. The Zynq die tracks this 3× worse (119 mHz rms), because it also follows the processor: it fell 0.76 °C within minutes of the log rolling over to part 2, while the TSic moved 0.13 °C.
- **Ramps cost 0.56 Hz per °C/min**; the temperature mode runs about 8 s ahead of the mass mode. That term explains the apparent 2.1 Hz loop between heating and cooling at 24–38 °C. With it removed, the two branches agree within 0.3 Hz above 28 °C. A real offset of 0.6 Hz (4 pm) remains after returning to 22.5 °C from 38.8 °C.
- **What is left is the cubic's shape:** 3.3 Hz peak to peak (0.60 Hz rms) for the step-free cubic once the drop, the board and the ramp term are removed. It sits mostly at 12–16 °C and 55–60 °C, between the points. Six points at only four distinct temperatures leave a cubic no redundancy: without the 12 °C point the curve moves by 36 Hz, and without the 38.8 °C point by 18 Hz.
- **T start matters, and the PT100 is good enough for it.** A T start that is 1 °C off costs 1.7 Hz at +5 °C from the start and 4.7 Hz at +20 °C. On slow stretches the PT100 agrees with the crystal to ±0.38 °C and lags it by only 20 s, so taking T start from the PT100 (the new setting) is the right default.

1 The run

The CSV is the two-part log as downloaded (part 2 begins at 18:59:52 CET when part 1 reached 64 MB). Everything below uses 10 s medians (2478 bins). Every compensated number is recomputed with the instrument's own `TempCompAlgorithm.FreqToTemp`, sample by sample, from the run's first values (T start = PT100 = 22.55 °C). Recomputing with the coefficients that were active during the run (the linear `infi2439_SC_1_wkcal`) reproduces the logged compensated column to 12 mHz. That profile showed 98.8 °C when the PT100 read 64.8 °C.

Item	Value
Samples	223 352 at 9.1 Hz, no gaps > 0.2 s; both PLLs locked 100 % of the time; no PT100 fault; no coefficient change mid-run
Temperature programme	22.6 → 38.8 → 22.5 → 11.5 → 64.8 °C, with a 2.4 h soak at 64–65 °C; fastest ramp +6.0 °C/min (16:20 CET), 80 % of the time slower than 0.2 °C/min
Mass mode fM	5.98185 MHz, span 166 Hz (turnover near 55 °C)
Temperature mode fT	6.566 MHz (lower B-region mode), span 8.3 kHz; amplitude 131 at 11.5 °C → 1356 at 64.8 °C
Board	TSic 31.46–32.48 °C; Zynq die 51.07–52.90 °C (first run with zynq_die_c logged)
Calibration points	16 taken during the run, six of them kept and fitted (ids 16–19, 22, 29), all auto-captured by the settle detector with the PT100 as source (180 s window, fT ≤ 10 Hz/min, PT100 ≤ 0.1 °C/min, 60 s hold). The last one was taken at 20:26 CET after the run stopped. Deleted before the fit: 14 (22.65 °C, the ambient start of the run), 15 (38.19 °C), 20 and 21 (added by hand before settling) and 23–28 (repeats at 64.5–64.8 °C). Fitted in the CALIBRATE tab and transferred to infi2439_SC_1 and infi2439_SC_1_wkcal at 20:26 CET

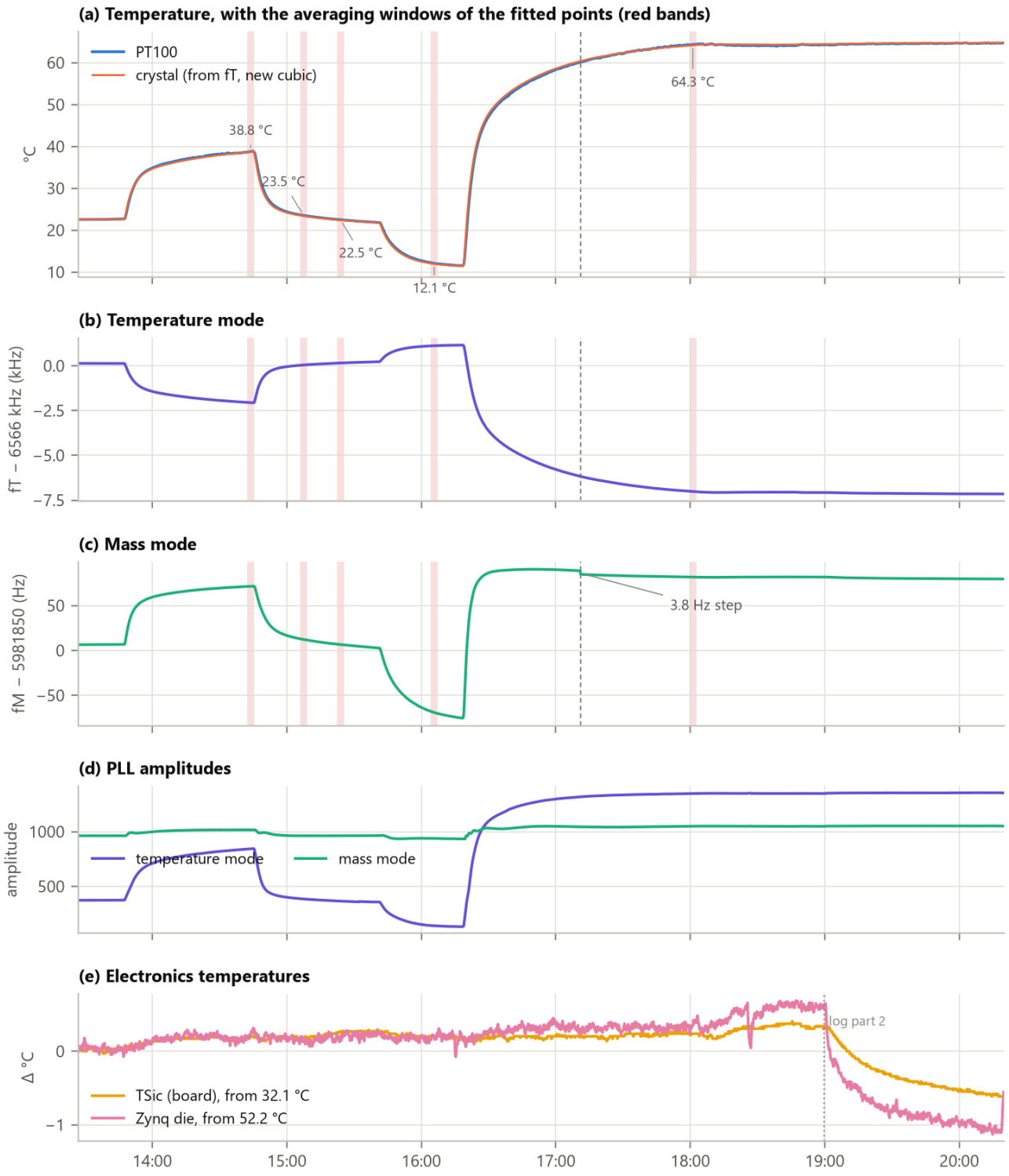


Figure 1. The run (CET). (a) PT100 and the crystal temperature solved from the two modes with your cubic. Red bands: the averaging windows of the five fitted points taken during the run (the 64.8 °C point came after the run stopped). Dashed line: the fM step. (b, c) The two modes. (d) PLL amplitudes: the temperature mode grows tenfold from 11.5 to 64.8 °C. (e) Board (TSic) and Zynq die temperature, relative to their start; the dotted line marks the log rollover.

2 The calibration

The CALIBRATE tab fits a cubic to fM(T) and fT(T) through the points, with the PT100 as T. Refitting the six stored points with numpy reproduces your coefficients to 0.03 % (the profile stores the browser's fit). Only the 1st- to 3rd-order terms are used: the algorithm takes the constant from the start frequencies.

Point	End (CET)	T (°C)	fT drift (Hz/min)	PT100 drift (°C/min)	Residual fM / fT (Hz)	Curve moves without it
16	14:45	38.77	−6.9	0.058	−0.05 / −1.9	18.1 Hz
17	15:09	23.51	9.4	−0.087	−0.13 / 3.0	0.2 Hz
18	15:25	22.48	4.9	−0.046	0.18 / −1.2	0.4 Hz
19	16:07	12.08	5.1	−0.087	−0.03 / −0.6	36.5 Hz
22	18:02	64.32	−8.9	0.067	0.64 / 21.5	0.5 Hz
29	20:26	64.76	−0.1	0.003	−0.62 / −20.8	0.6 Hz

Residuals are point minus cubic. The last column is the largest change in the run's compensated output when the cubic is refitted without that point.

Point quality is fine. The detector accepted points with fT still moving 5–9 Hz/min (0.05–0.09 °C/min), looser than the 2 Hz/min rule recommended earlier. On this Peltier set-up that is harmless: the PT100 lags the crystal by only 20 s, so the temperature error from the drift is under 0.03 °C, and the ramp term at 0.09 °C/min is 0.05 Hz. The two hot points differ by 141 Hz in fT for 0.43 °C of PT100 (330 Hz/°C against a slope of 226). During the long soak fT reported about 0.2 °C more warming than the PT100 did. That is a PT100-versus-crystal difference and does not touch the compensation, which uses no thermometer.

Coverage is the weakness. The points sit at four distinct temperatures (12.1, 22.5–23.5, 38.8, 64.3–64.8 °C), the same number as the cubic's coefficients. The fit therefore interpolates and nothing in it checks the curve between the clusters. Leaving out one of a near-duplicate pair changes little (0.2–0.6 Hz); leaving out 12.1 or 38.8 °C changes the run by 36 and 18 Hz.

PT100 range (°C)	dfT/dT measured	dfT/dT cubic	dfM/dT measured	dfM/dT cubic
11.5–13.0	−62.7	−74.1	10.00	8.75
21.5–24.0	−105.9	−110.5	5.53	5.83
35.0–38.5	−166.4	−156.9	3.08	2.53
60.0–65.0	−220.6	−226.4	−1.16	−1.09

Local slopes (Hz/°C) from straight-line fits to the slow stretches inside each range, against the cubic's derivative at their mean temperature.

The slopes agree with the cubic to 3–10 % except at the cold end, where the measured fM slope is 14 % steeper and fT 15 % shallower than the fit. That is the 12–16 °C error in Section 4.4. At 22 °C the temperature mode moves **−105.9 Hz/°C**. The weekend estimate from ambient swings (−91.2) was 14 % too shallow, within the uncertainty that report gave; this run supersedes it.

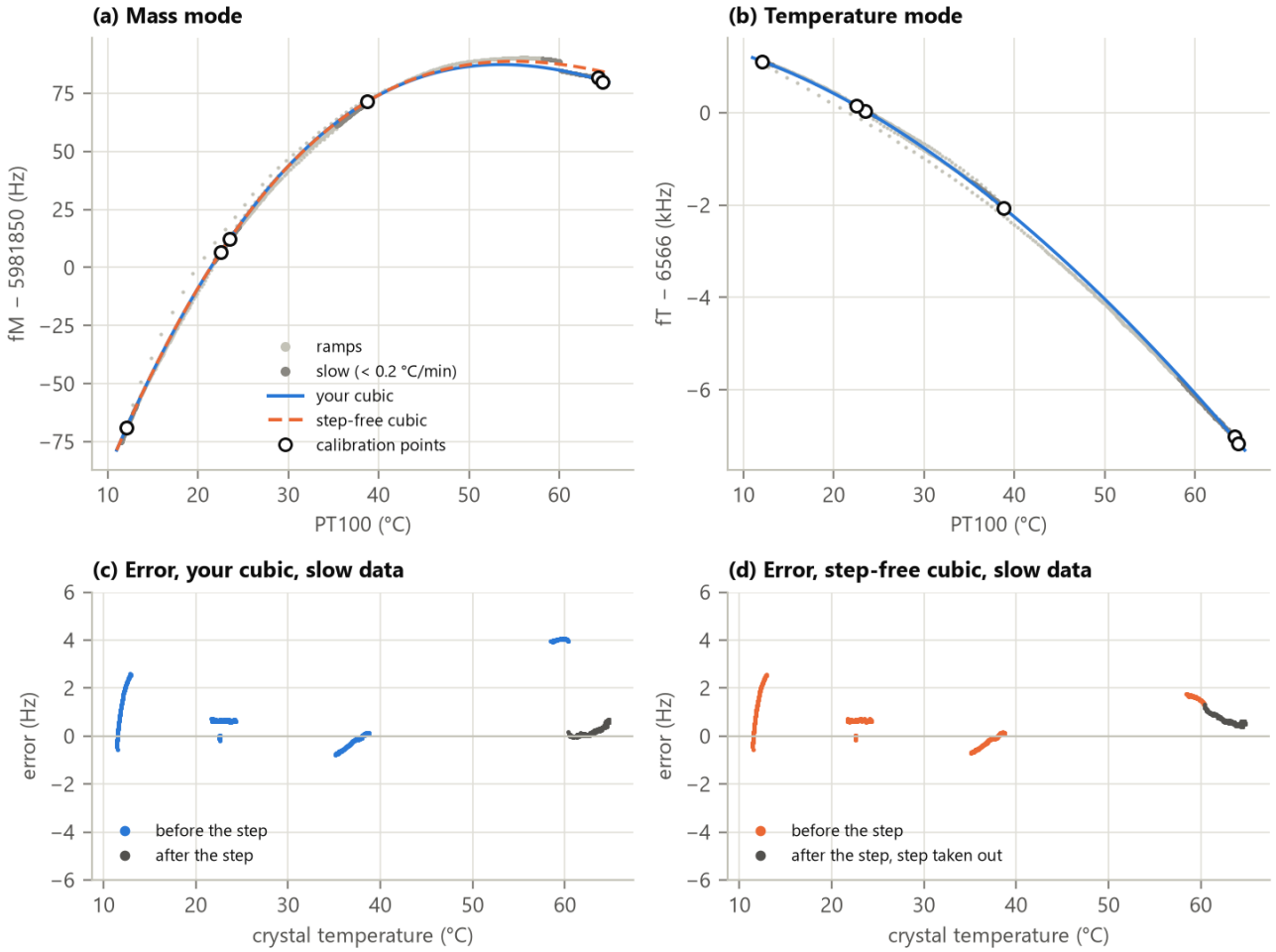


Figure 2. (a, b) Both modes against the PT100: the points, the slow data (grey), the ramps (light grey) and the two cubics. The ramp data fall off the curve in (b) because the PT100 lags the crystal by 20 s; the compensation does not use the PT100. (c, d) Compensated error against crystal temperature on the slow data, board temperature removed. With your cubic the data just before the step (58–60 °C) sit 4 Hz high. With the step-free cubic, and the step taken out of the later data, the hot end joins up.

3 How well it compensates

A run with no deposition should give a flat compensated output, so its excursions measure the error directly. The table removes the identified effects one at a time: the fM step (a constant after 17:11:01), the board temperature (1.98 Hz/°C × ΔTSic, from the hot soak) and the ramp term (−0.56 Hz per °C/min of crystal heating rate).

Compensated error (Hz)	Your cubic	Step-free cubic
Whole run, nothing removed	7.55 p2p · 1.23 rms	7.58 p2p · 2.21 rms
Slow stretches only (< 0.2 °C/min)	5.08 p2p	7.45 p2p
20–40 °C, nothing removed	4.79 p2p	4.74 p2p
Whole run, step and board removed	7.90 p2p · 2.02 rms	6.10 p2p · 0.72 rms
… and the ramp term	5.31 p2p · 1.95 rms	**3.33 p2p · 0.60 rms**

With nothing removed the two profiles score alike on this run. Your cubic was fitted to the hot points as they were after the drop, so it matches this run's hot end. Once the drop is taken out (it will not recur in the same place if it was a one-off), the step-free cubic is 3× better in rms. Below 40 °C the two are the same curve.

Metric	Value (step-free cubic, 11.5–64.8 °C)	Previous reference
M1 error band	3.3 Hz = 23 pm Au = 44 ng/cm² = 0.56 ppm (step, board and ramp term removed); 7.6 Hz as measured	1.26 Hz over 17.5–38 °C (TestCrystal3)
M2 gain	50× (22× as measured)	59×
M3 hysteresis	2.1 Hz loop at ~2 °C/min, of which all but ≤ 0.3 Hz is the ramp term; 0.6 Hz offset after the 38.8 °C excursion	0.70 Hz at 2 °C/min

M4 equivalent T error	1.1 °C (M1 ÷ the mean |dfM/dT| of 3.1 Hz/°C)	0.35 °C
M5 drift at constant T	0.02 Hz/h at 64 °C with the board removed (−0.92 Hz/h with it)	−0.42 Hz/day

M1–M5 as defined on 10 Aug. The reference crystal covered 20 °C; this run covers 53 °C, including the SC-cut's turnover near 55 °C, where fM barely moves and M4 says little.



Figure 3. (a) The compensated output your cubic gives for this run. (b) The step-free cubic with the step and the board temperature removed (orange), and with the ramp term removed too (black). From 18:00 to the end the black line is flat to ±0.1 Hz, for more than two hours. (c) Crystal heating rate from the solved temperature.

4 What is not compensated

4.1 A 3.8 Hz drop in fM alone

At 17:11:01 CET, heating through 60.1 °C at 0.15 °C/min, fM fell 3.80 Hz between two samples (0.11 s apart) and then carried on as before. fT changed by 0.13 Hz, inside its noise, and neither amplitude nor phase moved (Figure 4). That rules out the obvious causes:

- **temperature**: the crystal cannot change temperature in 0.1 s, and fT would have moved by 200 Hz per °C;
- **the clock**: fT would have moved 4.2 Hz with it;
- **uniform mass loading**: fT would have fallen 4.2 Hz too;
- **an activity dip**: the amplitude would have dipped.

What remains is something that couples to the C-mode and not to the B-region mode: a mounting or contact shift (the C-mode's mode shape and its stress sensitivity differ from the B-mode's), or a small particle where only the C-mode has energy. Nothing in this run tells which, or whether it reverses on cooling. It is worth 26 pm Au, and the

compensation passes all of it through as mass.

What it did to the calibration. Points 22 and 29 were taken after the drop and the rest before, so the cubic bent its upper part down to meet them. If the step-free curve is the crystal's real behaviour, your cubic is off by 0.3 Hz at 45 °C, 1.5 at 55 °C and 3.8 at 64.5 °C (start 22.5 °C), and by less than 0.1 Hz anywhere below 40 °C. The step-free profile is your six points with 3.80 Hz added to the two hot fM values. If the drop was a one-off, it is right everywhere. If the drop reverses on cooling, it is still right below 60 °C, and no smooth cubic can follow the jump anyway.

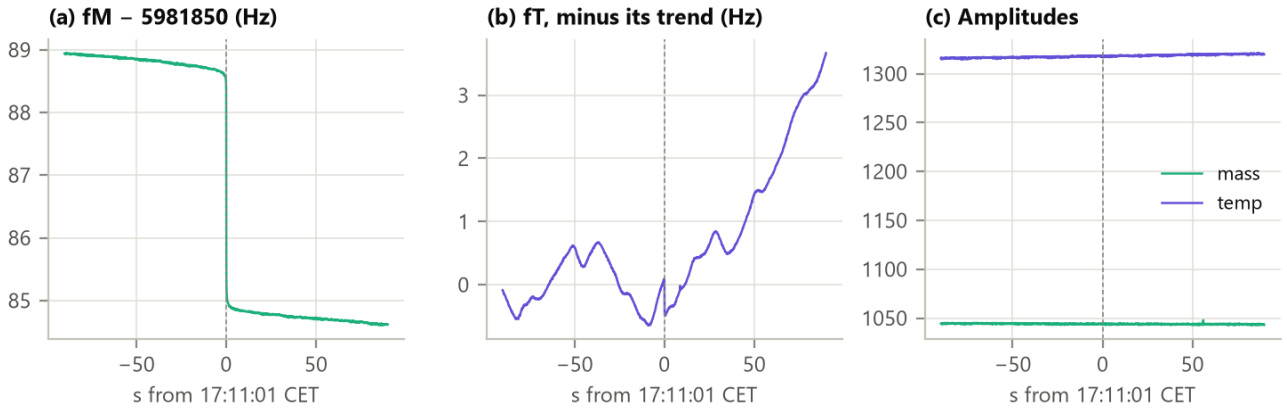


Figure 4. Every sample in the three minutes around the step. (a) fM. (b) fT minus a straight line fitted to the 90 s before: no step. (c) Amplitudes: unchanged.

4.2 Ramp rate, and heating against cooling

During ramps the error moves with the heating rate: −0.56 Hz per °C/min (fitted over the whole run with a smooth function of temperature alongside). It can also be written as a time offset: the temperature mode leads the mass mode by 8 s, which fits equally well (0.44 against 0.46 Hz rms). The B-region modes of an SC-cut are not stress-compensated, so a temperature gradient across a heating crystal shifts them first. At 1 °C/min this is 0.56 Hz (3.9 pm). Heating the crystal during a deposition therefore makes the film read slightly thick while it heats (thin while it cools), and the error disappears once the temperature is steady again.

Figure 5 shows the excursion to 38.8 °C. Raw, the heating and cooling branches are 2.1 Hz apart, which looks like hysteresis. With the ramp term removed they fall on top of each other above 28 °C (within 0.3 Hz), and what is left is the curve error the two branches share. Back at 22.5 °C an offset of 0.64 Hz remains with the board removed (1.11 Hz raw). That is a 4.4 pm apparent mass loss after one 16 °C excursion, from thermal hysteresis or desorption; one excursion cannot separate the two. **Answer to the open question:** data taken during slow ramps fall on the plateau curve. Below 0.2 °C/min the ramp term is under 0.12 Hz, so a slow continuous ramp in both directions can replace soaks.

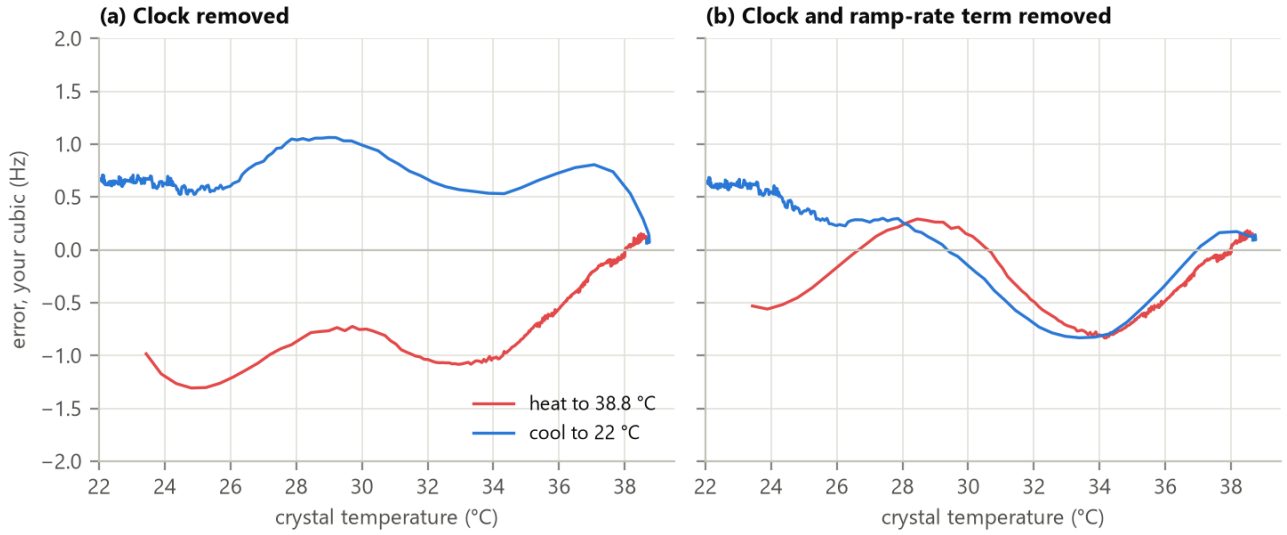


Figure 5. Error of your cubic against crystal temperature during the heating to 38.8 °C (red) and the cooling back (blue). (a) With the board temperature removed. (b) With the ramp term also removed.

4.3 Board temperature: the clock, and which thermometer to use

The 64 °C soak is the cleanest test of the electronics because the crystal hardly moves. At 19:00 CET the board started to cool (0.92 °C over the following hour) and the compensated output fell 1.75 Hz with it. Regressed on the solved temperature and the TSic, it leaves 40 mHz rms (against 152 mHz without the TSic) at **1.98 Hz/°C = 0.33 ppm/°C**. Fitted to each mode separately, fT moves 1.16× as much as fM. That is close to the 1.098 a reference-clock error gives (both modes shift by the same fraction), and it repeats the weekend's finding: 0.25–0.4 ppm/°C, ratio 0.76–1.31. Over the whole run the same term gives 2.23 Hz/°C.

The Zynq die, logged for the first time in this run, does worse: 119 mHz rms at its best and 51 mHz after an 8 min lag is applied, against 40 mHz for the TSic without any lag. The die responds to the processor as well as to the board. Within three minutes of the log rolling over to part 2 it fell 0.76 °C while the TSic moved 0.13 °C (Figure 6b). At 18:26 CET it dipped half a degree for two minutes, and the error dipped 0.15 Hz with it, so it does see some fast events that the TSic misses. For a clock correction, use the TSic. Why the die's load changes with the log is not traced; the die warmed by 0.6 °C during the last hour of part 1.

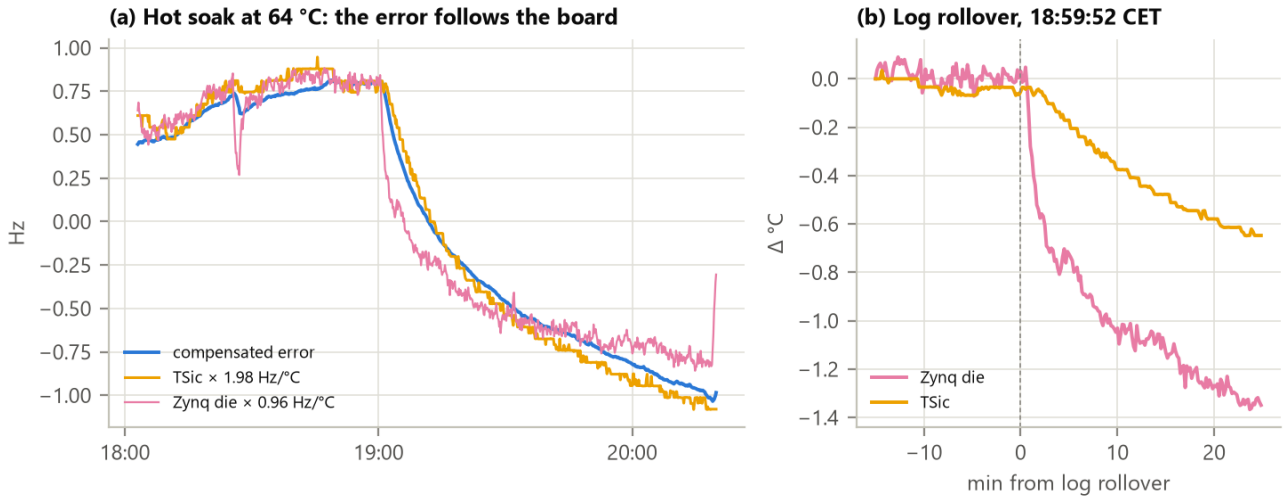


Figure 6. (a) During the 64 °C soak the compensated error (blue) follows the TSic scaled by 1.98 Hz/°C (orange); the Zynq die (pink) follows less well. (b) Die and TSic around the log rollover.

4.4 The shape of the cubic

With the step, the board and the ramp term out, the step-free cubic leaves 3.3 Hz peak to peak. The error sits mostly between points. It reaches +2.6 Hz at 13–15 °C on the way down to 12 °C, where the measured slope is steeper than the fit, and +1.7 Hz at 55–60 °C. More than half of this run's slow data are at 64 °C and the rest at four places, so the best cubic this run supports is no better (3.1 Hz on the slow data). A 5th-order curve would reach 1.8 Hz, but the instrument uses a cubic. The way to do better is more temperatures, not more points per temperature: see Section 5.

4.5 T start

The compensation anchors the curve at T start. If T start is off by δ, the curvature of the two modes, the temperature mode's above all (−63 Hz/°C at 12 °C, −221 Hz/°C at 64 °C), turns δ into an error that grows with distance from the start (Figure 7). Per °C of δ:

Run	+2.5 °C	+5 °C	+10 °C	+20 °C	to 12 °C	to 64.5 °C
starts at 22.5 °C	0.9 Hz	1.7 Hz	3.0 Hz	4.7 Hz	−6.0 Hz	6.3 Hz

A deposition that heats the crystal by 5 °C therefore loses 1.7 Hz (12 pm) per degree of T start error. Typing 22.5 °C when the crystal is really at 25 °C costs about 4 Hz at +5 °C. On slow stretches the PT100 agrees with the solved crystal temperature to −0.34…+0.38 °C (0.16 °C rms), or about 0.3 Hz typically (0.6 Hz at worst) at +5 °C. Set T Start Source to PT100 (a333f10) once that is deployed.

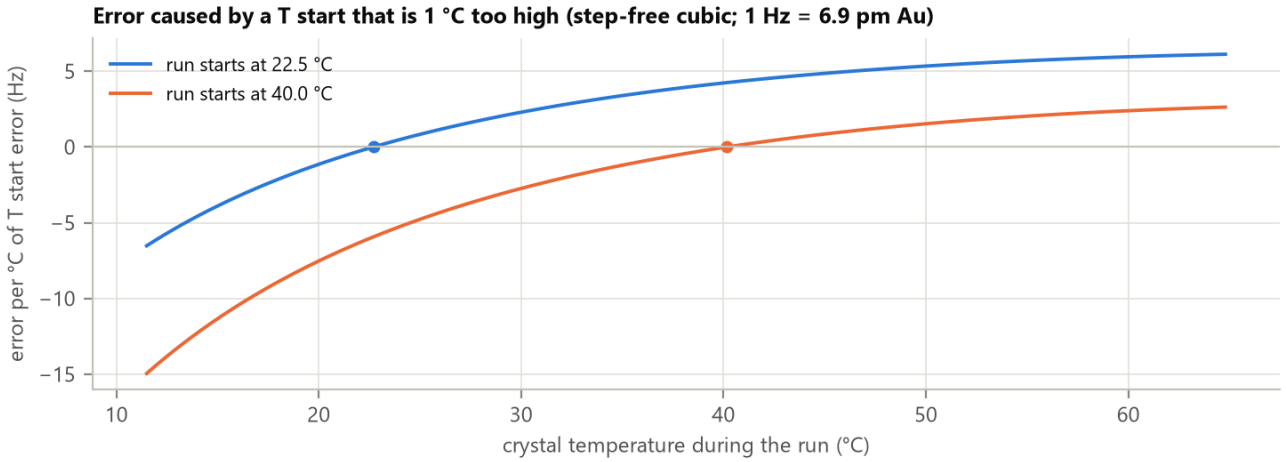


Figure 7. Error per °C of T start error, against the temperature the crystal reaches, for runs starting at 22.5 and 40 °C (step-free cubic; your cubic is the same below 40 °C).

4.6 Noise and the temperature mode

Noise is negligible: after removing a 5 min running mean, the 10 s medians of the compensated output scatter by 8 mHz at 22.5 °C, 6 mHz at 11.7 °C and 4 mHz at 64.7 °C. The temperature mode's amplitude rose from 131 at 11.5 °C to 1356 at 64.8 °C, as intended when the lower mode was chosen, and the lock held throughout. Its weakest point is now the cold end. Below about 10 °C it would approach 100; check the lock-detect threshold before calibrating colder.

5 Recommendations

- **Use the step-free profile** (`data/crystals/infi2439_SC_1_stepfree.json`, Appendix B), or keep yours if you stay below 40 °C: the two are the same there. Upload it under its own name or copy its six coefficients into infi2439_SC_1, between runs, not during one.
- **Find out whether the drop reverses.** With the step-free profile active, start a run at 64 °C (T start from the PT100) and cool slowly to 22 °C. If the output stays flat, the drop was permanent and the step-free profile is right everywhere. If it jumps by about +4 Hz somewhere on the way down, the drop reverses, and the temperature where it does so is worth knowing for depositions. Either way, fM against fT near 60 °C on a second pass shows whether it recurs.
- **Next calibration: more temperatures, both directions.** Six to eight distinct plateaus (for example 12, 17, 22, 30, 38, 46, 54, 62 °C) give the cubic something to average over and expose a bad point. Points 17/18 and 22/29 were near-duplicates and added little. A slow continuous ramp (≤ 0.2 °C/min) both ways also works and gives the curve between plateaus, as Section 4.2 shows.
- **Correct the clock from the TSic** if long runs matter: fM_corr = fM − 1.98 Hz/°C × (T_TSic − T_TSic,start), and fT by 1.098× that. It is worth 14 pm Au per °C of board temperature. For a few-minute deposition with a steady board it is below 1 pm. An external reference clock removes it at the source.
- **Keep the settle detector's 10 Hz/min** on this set-up: with a 20 s PT100 lag it costs nothing measurable. The 2 Hz/min rule came from the slower ambient set-up.
- **Housekeeping.** infi2439_SC_1_wkcal was the active profile during the run, and its total_deposited rose by 3.42 nm. That is the linear profile's error at 64 °C, not a film. On the Pitaya both infi2439_SC_1 and infi2439_SC_1_wkcal now hold your cubic.

Appendix A Method details

- **Binning.** 10 s medians of every channel. The step and the rollover are timed on the raw samples.
- **Compensated error.** TempCompAlgorithm.FreqToTemp from the repo, run over the 10 s series from the first bin (fM, fT, T start = 22.549 °C), minus the first value. No mass was deposited, so the output is the error.
- **Step.** Straight lines fitted to the 60 s before and after 17:11:01.300 CET at full rate, compared at the step time.
- **Board coefficient.** Hot soak only (after 18:02 CET): error = a + b·(T − 64.5) + c·ΔTSic. Lagged versions of both sensors scanned 0–60 min in 2 min steps; the TSic is best unlagged. The fT/fM ratio comes from the same fit applied to each mode.
- **Ramp term.** The heating rate is the derivative of the solved temperature after a 60 s low-pass. It is fitted over the whole run together with a cubic in temperature that absorbs the curve error, after the step and board terms are removed.
- **Hysteresis.** Mean error within ±0.25 °C of each 2 °C step on the heating (13:48–14:43) and cooling (14:43–15:42) branches.
- **PT100 lag.** rms of PT100 − lowpass(solved temperature, τ), scanned 0–120 s in 5 s steps.
- **Leave-one-out.** Cubic refitted to five points, run recomputed, largest difference from your cubic's output.
- **T start.** Synthetic run along the step-free curve, compensated with T start + 1 °C, error read at each temperature.
- **Profile check.** The JSON in Appendix B, loaded through the repo's CrystalManager, reproduces the step-free evaluation to 0.0 µHz.
- **Times** are CET (UTC+1). The fitted points are from data/calibration/points.json on the Pitaya (ids 16, 17, 18, 19, 22, 29); refitting them reproduces your coefficients to 0.03 %. Each one's fM, fT and PT100 match the run log averaged over its stored window to within 1 mHz and 1 mK. The deleted points (14, 15, 20, 21, 23–28) come from the service log. Adding point 14, the ambient start, to the fit would change this run's compensated output by at most 0.54 Hz.

Appendix B Profile file


```
{
  "name": "infi2439_SC_1_stepfree",
  "freq_mass": 5981853.0,
  "freq_temp": 6566157.0,
  "freq_virgin": 6000000.0,
  "fM_0": 5981652.267825026,
  "fM_1": 12.681585697394432,
  "fM_2": -0.17698442705986733,
  "fM_3": 0.0007478333923259184,
  "fT_0": 6567755.730664034,
  "fT_1": -30.249435524132693,
  "fT_2": -1.9056392098316415,
  "fT_3": 0.0038903973408308434,
  "mass_sensitivity": -1.33e-07,
  "sens_area": 5.25e-05,
  "tooling_ratio": 1.0,
  "hours_active": 75.48217099222873,
  "total_deposited": 0.19739988851187762,
  "created": "2026-11-03T08:21:10.192440+00:00",
  "last_modified": "2026-11-03T08:21:10.192440+00:00"
}
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